Ans. (a) Let the time taken to walk = x
and the time taken to drive = y
According to question,

$x + y = 6$ _____(i)

and $x + x = 10$

$2x = 10$

$x = 5$

Putting the value of x in eq^n (i)

$\because x + y = 6$

$5 + y = 6$

$y = 1$

time taken by the person to drive both way

$\Rightarrow y + y = 2y$

$= 2 \times 1 \quad \{\because y = 1\}$

$= 2$ hr

63. Four cars are hired at the rate of Rs. 6 per km. and the cost of diesel at Rs. 40 a litre. In this context, consider the details given in the following table :

Car	Mileage (Km/L)	Hours	Total Payment (Rs.)
A	8	20	2120
B	10	25	1950
C	9	24	2064
D	11	22	1812

Which car maintained the maximum average speed ?

(a) Car A (b) Car B
(c) Car C (d) Car D

Ans. (a) $\because$ four cars are hired at the rate of Rs. 6 per km.

Distance covered by Car A $= \frac{2120}{6} = 353.33$ km.

Distance covered by Car B $= \frac{1950}{6} = 325$ km.

Distance covered by car C $= \frac{2064}{6} = 344$ km.

Distance covered by Car D $= \frac{1812}{6} = 302$ km.

$\because$ Average speed $= \frac{\text{total distance}}{\text{total time}}$

The average speed of Car A $= \frac{353.33}{20} = 17.66$ km/hr

The average speed of Car B $= \frac{325}{25} = 13$km/hr

The average speed of car C $= \frac{344}{24}$

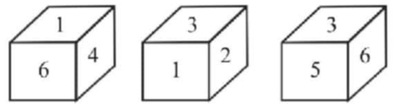
$\Rightarrow$ 14.33 km.

The average speed of Car D $= \frac{302}{22}$

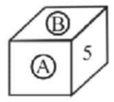
$= 13.72$ km

Hence Car A maintained the maximum average speed.

64. A cube has six number marked 1, 2, 3,4 , 5 and 6 on its faces. Three views of the cube are shown below



What possible numbers can exist on the two faces marked (A) and (B), respectively on the cube ?



(a) 2 and 3 (b) 6 and 1
(c) 1 and 4 (d) 3 and1

Ans. (a) Considering three views of the cube,

$2 \xrightarrow{\text{opposite}} 6$

$1 \xrightarrow{\text{opposite}} 5$

$3 \xrightarrow{\text{opposite}} 4$

$\because$ The opposite of 5 is 1.
$\therefore$ Neighbour of 5 will not be 1
Hence, 2 and 3 numbers can exist on the two faces marked (A) and (B), respectively on the cube.

Directions for the following 5 (five) items –

Study the two figures given below and answer the five items that follow :

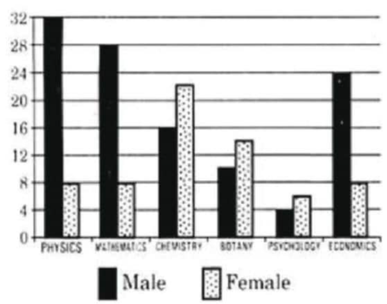
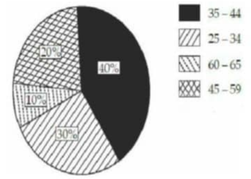


Figure 1 : Number of Professors in selected disciplines in a University by sex



65. How many Physics professors belong to the age group 35-44 ?

(a) 18 (b) 16
(c) 14 (d) 12

Ans. (b) The Number of Physics teacher

$= (32 + 8) \times \frac{40}{100} = 40 \times \frac{40}{100} = 16$

66. Which one of the following disciplines has the highest ratio of males to females ?

(a) Physics (b) Mathematics
(c) Chemistry (d) Economics

Ans. (a) The ratio of males to females,

In physics,

$M = F = 32 : 8$

$= 4 : 1 \Rightarrow \frac{4}{1} = 4$

In mathematics,

$M = F = 28 : 8$

$= 7 : 2 \Rightarrow \frac{7}{2} = 3.5$

In chemistry,

$M : F = 16 : 22$

$= 8 : 11 \Rightarrow \frac{8}{11} = 0.72$

In economics,

$M : F = 24 : 8$

$= 3 : 1 \Rightarrow \frac{3}{1} = 3$

67. What percentage of all Psychology professors are females?

(a) 40% (b) 50%
(c) 60% (d) 70%

Ans. (c) Required % $= \frac{6}{(6+4)} \times 100\%$

$= \frac{6}{10} \times 100\% = 60\%$

68. If the number of female Physical professors in the age group 25-34 equals 25% of all the Physical professors in the age group, then what is the number of male Physics professors in the age group 25-34 ?

(a) 9 (b) 6 (c) 3 (d) 2

Ans. (a) The number of all physical professors $= 32 + 8$

$= 40$

The number of all physical professors which age group (25 – 34) $= 40 \times \frac{30}{100}$

$= 12$

The number of female physical professors which age group (25 – 34) $= 12 \times \frac{25}{100}$

$= 3$

Then, the number of male physical professors which age group (25 – 34) $= 12 - 3$

$= 9$

69. If the Psychology professors in the University constitute 2% of all the professors in the University, then what is the number of professors in the University ?

(a) 400 (b) 500
(c) 600 (d) 700

Ans. (b) Let the number of professors in the university = x

According to question,

$x \times 2\% = (6 + 4)$

$x \times \frac{2}{100} = 10$

$x = 500$

70. Four friends, A, B, C and D distribute some money among themselves in such a manner that A gets one less than B, C gets 5 more than D, D gets 3 more than B. Who gets the smallest amount ?

(a) A (b) B
(c) C (d) D

Ans. (a) Let $A = x$

$B = (x + 1)$

$C = (x + 4) + 5 \Rightarrow (x + 9)$

$D = (x + 1) + 3$

Hence, A gets the smallest amount.